Customer Segmentation & Service Recommendation System

Veda Technology — Data Science Internship (Day 1 of 4)

Project overview

This report demonstrates an end-to-end customer analytics workflow using synthetic data: cleaning, exploratory analysis, feature engineering, scaling, K-Means clustering, PCA visualization, segment profiles, and explainable recommendations.

Business problem and objectives

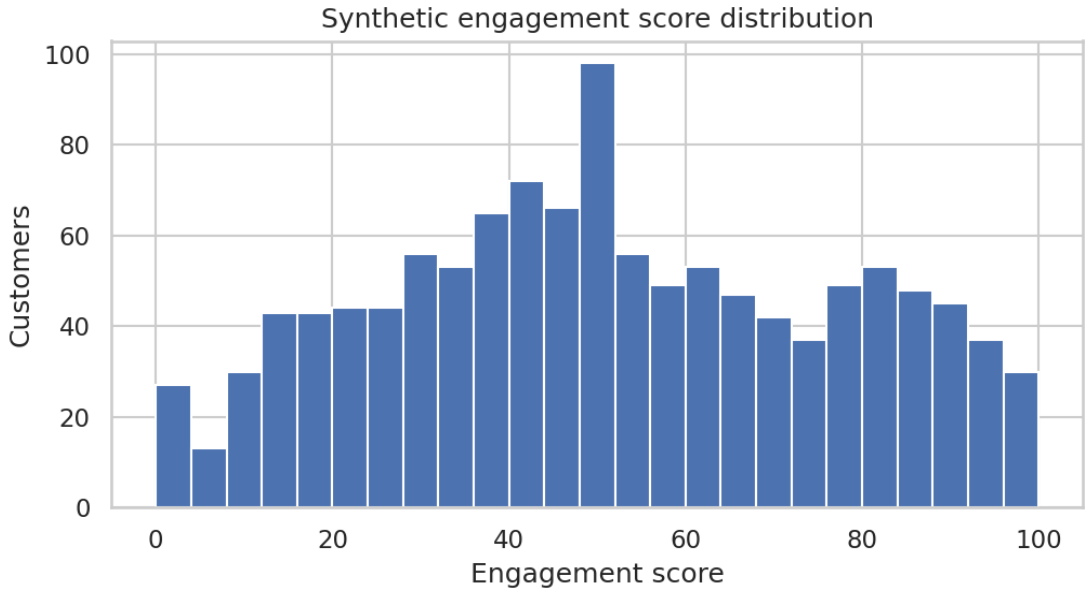
Customers may differ in engagement, website visits, service inquiries, purchasing, and support needs. Behavioral segments can help tailor learning resources and service journeys. This educational project generates 1,200 synthetic customer records and does not use private or confidential company data.

Dataset and cleaning

The raw synthetic dataset contains 1208 rows including intentionally duplicated records and missing values. After exact duplicate removal and median/mode imputation, the cleaned dataset contains 1200 rows and 12 original fields. Numeric missing values are filled with medians; categorical missing values use the mode.

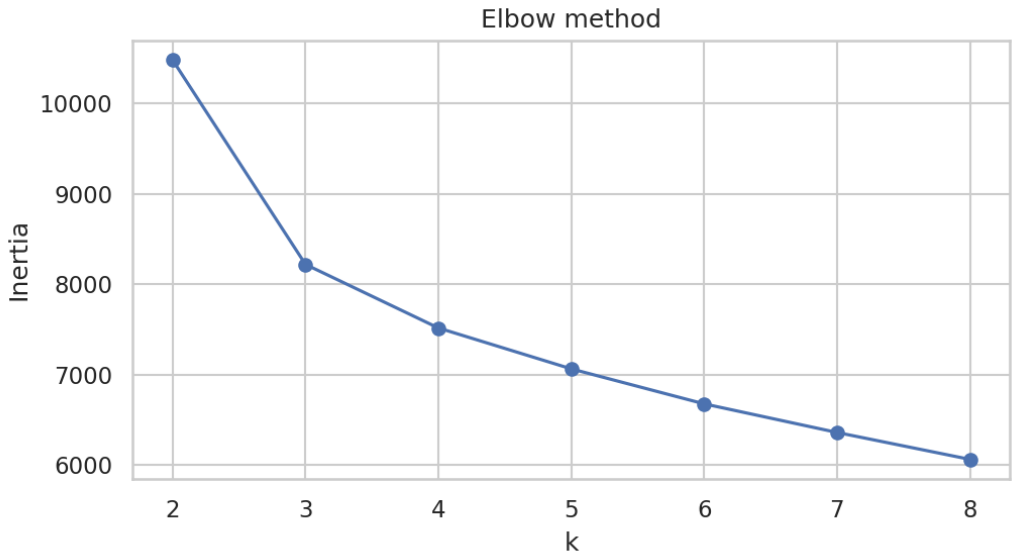
Fields include age, customer type, preferred program, website visits, session duration, engagement, service inquiries, purchases, spending, support interactions, and satisfaction.

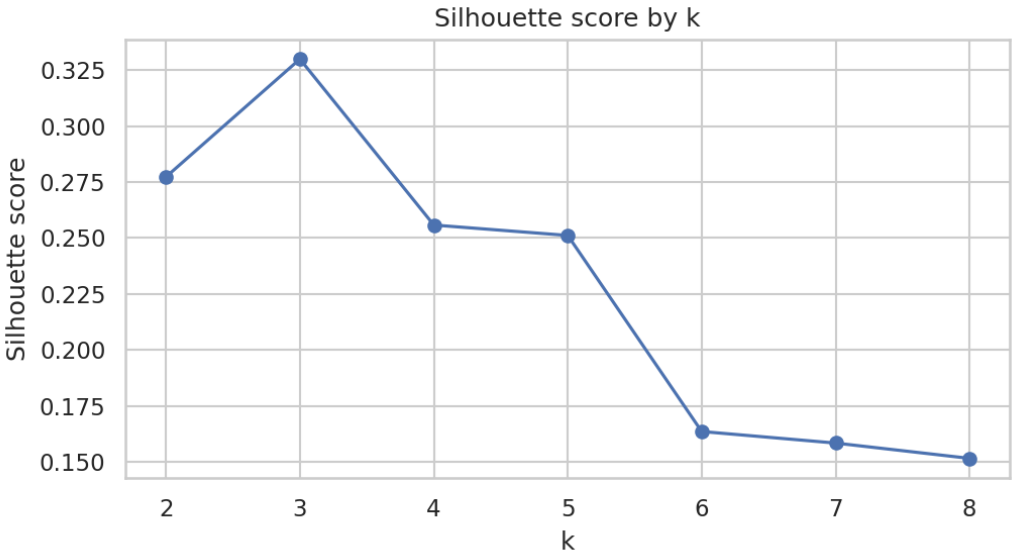
Exploratory analysis



Feature engineering and methodology

Derived features are visits per purchase, inquiry rate per visit, and engagement per visit. Customer_ID is excluded from clustering. StandardScaler is applied because K-Means uses distance and numeric fields have different scales. Candidate k values from 2 to 8 are compared using inertia and silhouette score.

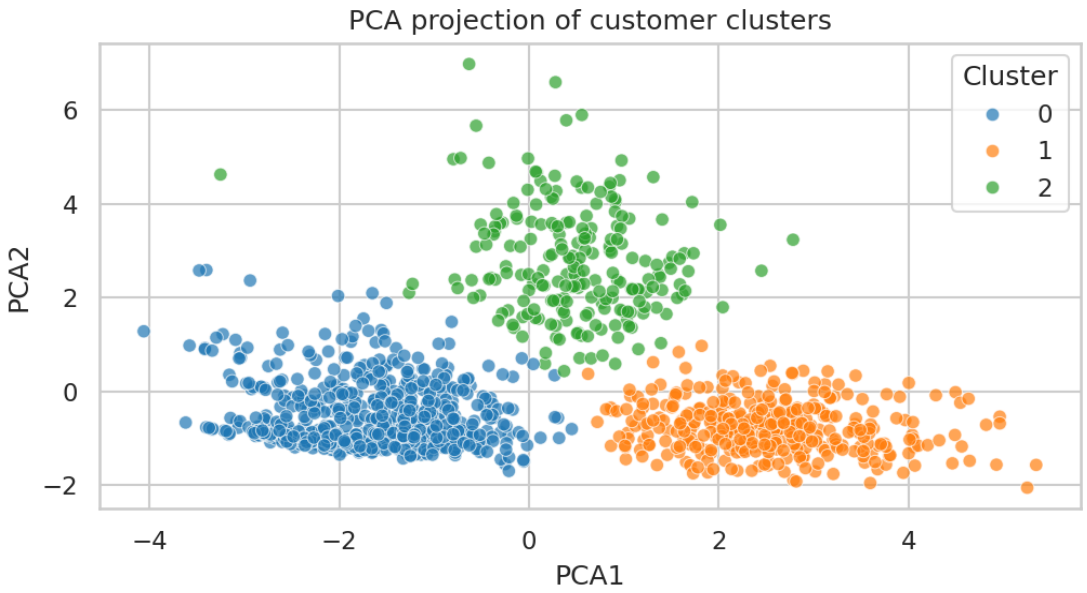




Calculated evaluation results
The candidate with the highest silhouette score among k=2..8 is k=3. Its silhouette score is 0.3299. This value was calculated from the synthetic dataset in this report-generation run. Review the elbow curve and cluster profiles too; no single score establishes business usefulness.

PCA visualization

The first two principal components explain 52.7% of standardized feature variance in this run. PCA is used for visualization; clustering was performed on the standardized original engineered features.


PCA projection of customer clusters
Cluster
0
1
2
PCA2
PCA1
6
4
2
0
−2
−4
−2
0
2
4

Customer segment profiles

Names below are heuristic summaries assigned from measured cluster means. Cluster IDs are arbitrary. A segment may share a descriptive label with another cluster if its profile meets the same rules, so examine numeric profiles alongside labels.

Cluster	Segment name	Customers	%	Visits	Engagement	Inquiries	Purchases	Satisfaction
0	New or Low-Engagement Customers	619	51.6	3.1	32.6	0.7	0.6	3.68
1	Highly Engaged Customers	365	30.4	17.8	81.0	1.9	3.0	4.37
2	Highly Engaged Customers	216	18.0	8.6	54.2	7.4	1.2	3.69

Recommendation methodology

Recommendations use transparent rules: highly engaged customers are offered advanced mentoring/pathways; service-focused customers are offered consultation and onboarding; low-engagement customers are offered beginner orientation; occasional visitors receive a targeted webinar or intermediate pathway. Low satisfaction adds a follow-up to understand concerns before promotional outreach. Preferred program personalizes the recommended learning program.

Example recommendations

Customer_ID	Segment_Name	Recommended_Program	Recommended_Service
CUST-00001	New or Low-Engagement Customers	Data Science	Beginner orientation, starter resources, and introductory workshop
CUST-00002	New or Low-Engagement Customers	Generative AI	Beginner orientation, starter resources, and introductory workshop
CUST-00003	Highly Engaged Customers	Data Science	Advanced project mentoring or advanced learning pathway + satisfaction follow-up
CUST-00004	New or Low-Engagement Customers	Generative AI	Beginner orientation, starter resources, and introductory workshop
CUST-00005	Highly Engaged Customers	Generative AI	Advanced project mentoring or advanced learning pathway

Business insights and recommendations

Use actual segment profiles to compare engagement, visits, spending, inquiries, and satisfaction. Pilot recommendations with opt-in customers, measure enrollment/conversion and satisfaction, track opt-outs, and revise rules based on outcomes. Treat low satisfaction as a reason for helpful follow-up rather than immediate promotion.

Limitations

All records are synthetic. K-Means is sensitive to scaling, outliers, and k and tends to favor compact, roughly spherical clusters. Silhouette score is one diagnostic, not a business success measure. Segment labels are heuristic; recommendations are not validated against real outcomes. The model uses numerical features, while categorical preferences are used for interpretation and personalization.

Conclusion

The project illustrates a reproducible end-to-end clustering workflow and exports customer-level segment assignments, profile summaries, evaluation metrics, and rule-based recommendations. Before operational use, validate the workflow with authorized representative data, review fairness and privacy, and measure outcomes in a controlled pilot.

Generated report includes calculated synthetic-data metrics from this run. Do not interpret these results as real Veda Technology customer findings.